强化学习baseline技术文档

本文档介绍了一个基于 NVIDIA Isaac Gym 的强化学习环境。Pi_rl_baseline 是一个面向高擎机电双足机器人 Mini Pi 的 sim2sim 框架，支持从 Isaac Gym 到 Mujoco 的策略迁移。该框架使用户能够在不同的物理引擎中验证训练得到的策略，从而评估其稳健性与泛化能力。

建议结合高擎开源项目 https://github.com/HighTorque-Robotics/livelybot_pi_rl_baseline 的 README 一同学习，以获得更完整的理解。

以下是高擎机电强化学习入门教学

适合用于向初学者展示如何基于仿真环境训练和迁移策略。

1. 自行安装**miniconda或Anaconda；**

2. 创建一个虚拟环境：

```
conda create -n pi_env python=3.8
```

3. 查看Nvidia显卡驱动：

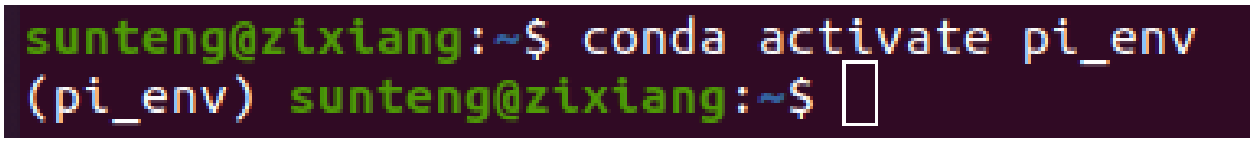
在命令行中使用命令 `nvidia-smi` 以查看驱动的CUDA版本，可以看到CUDA版本为12.8，驱动版本为570

```
sunteng@zixiang:~$ nvidia-smi
Tue Jun  3 13:04:55 2025
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 570.133.20             Driver Version: 570.133.20     CUDA Version: 12.8     |
|-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  NVIDIA GeForce RTX 4060 Ti     Off |   00000000:01:00.0  On |                  N/A |
|  0%   42C    P8             12W /  165W |    1067MiB /  16380MiB |     23%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                      GPU Memory |
|        ID   ID                                                             Usage      |
|=========================================================================================|
|    0   N/A  N/A            1380      G   /usr/lib/xorg/Xorg                      59MiB |
|    0   N/A  N/A            2194      G   /usr/lib/xorg/Xorg                     522MiB |
|    0   N/A  N/A            2330      G   /usr/bin/gnome-shell                    86MiB |
|    0   N/A  N/A            2563      G   .../sunloginclient --cmd=autorun        12MiB |
|    0   N/A  N/A            2707      G   ...me/58.0.3029.81 Safari/537.36         2MiB |
|    0   N/A  N/A            2733      G   ...D4C2B51A7D651DD79862114E7657D         6MiB |
|    0   N/A  N/A            3852      G   /usr/lib/firefox/firefox               193MiB |
|    0   N/A  N/A          346176      G   ...OTP --variations-seed-version        31MiB |
+-----------------------------------------------------------------------------------------+
```

4. 激活创建的conda环境：

```
conda activate pi_env
```



5. 安装PyTorch：

```
conda install pytorch torchvision torchaudio pytorch-cuda=12.4 -c pytorch -c nvidia
```

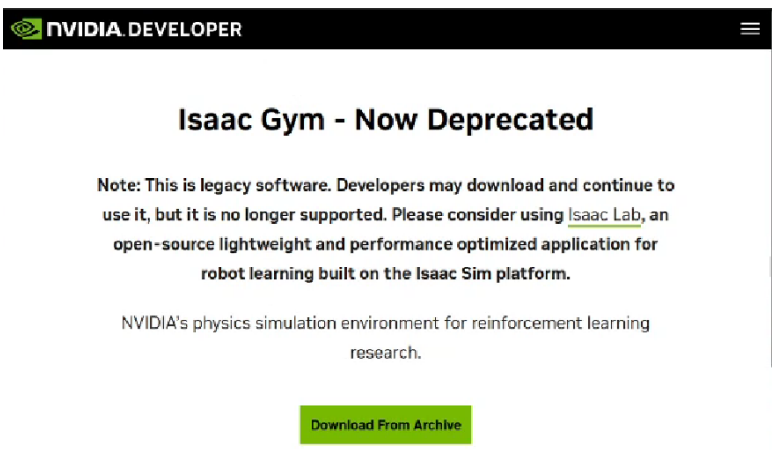
（所选择的 `CUDA` 版本要小于等于电脑安装的版本，这里选择的是CUDA 12.4）

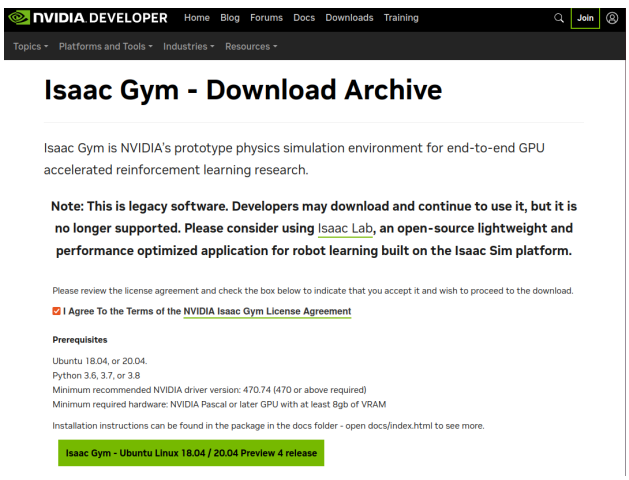
6. 使用conda安装numpy：

```
conda install numpy=1.23
```

7. 安装 `Isaac Gym`：

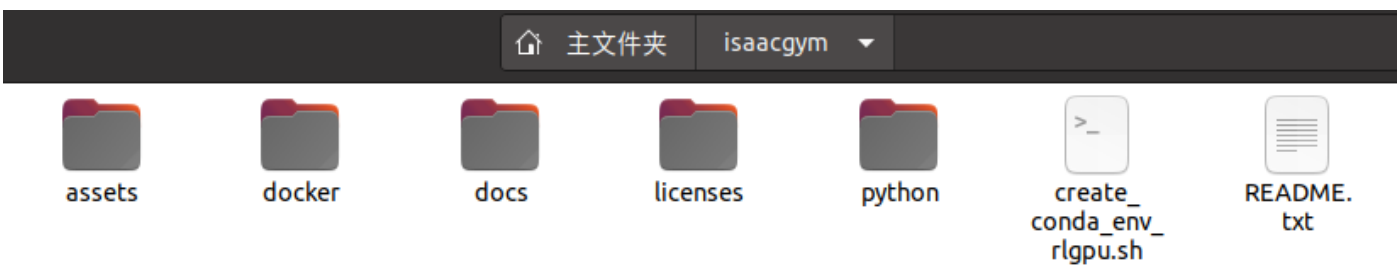
在Nvidia官网下载并安装 Isaac Gym Preview 4 *（注意：仅支持Ubuntu18.04或20.04）*





8. 提取到主目录：

将下载下来的Isaac_Gym_Preview_4_package提取到主目录下，并将子文件夹下的isaacgym包剪切到主目录下；



9. 进入 isaacgym的包中进行安装：

- 先输入 `cd isaacgym/python`
- 再输入 `pip install -e .`

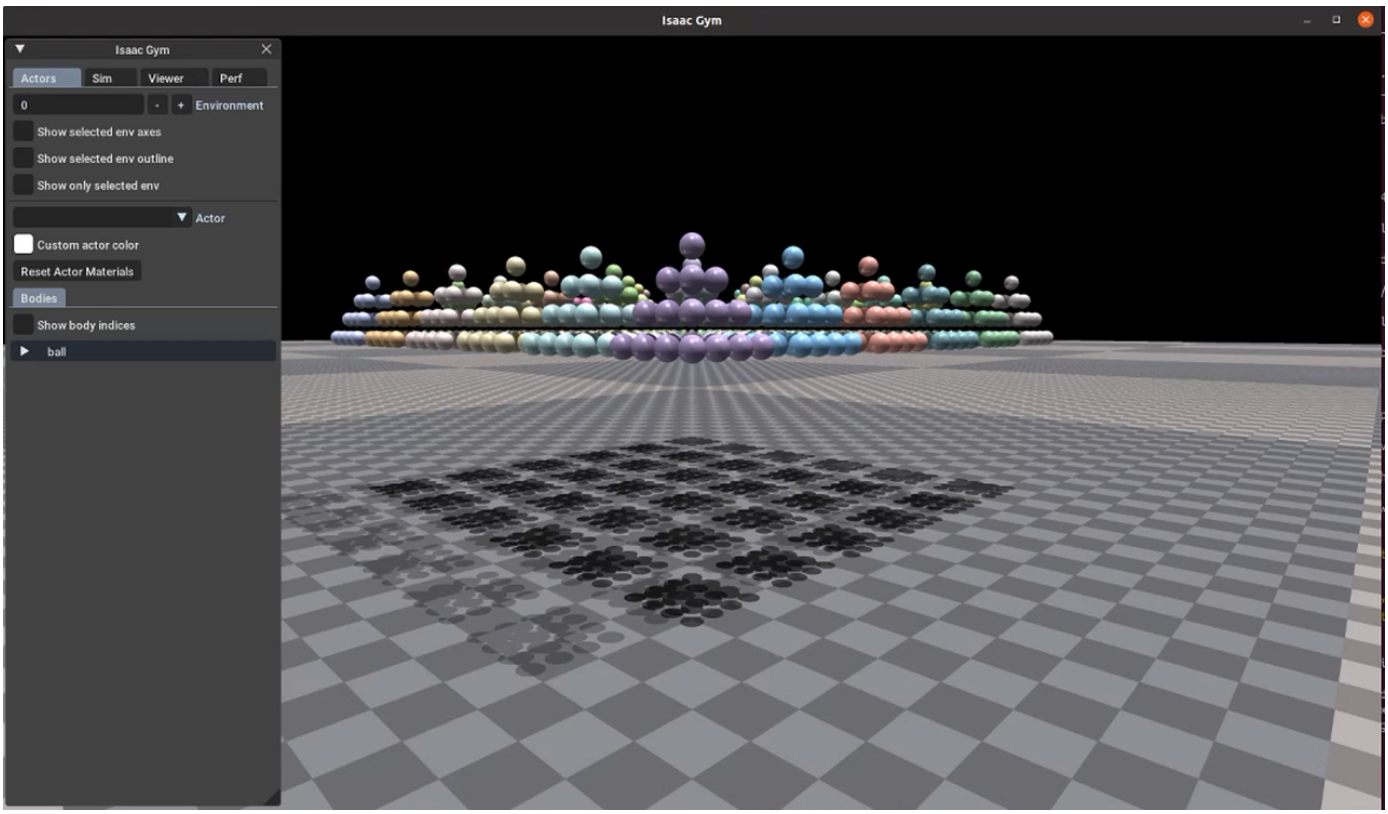
```
(pi_env) sunteng@zixiang:~$ cd isaacgym/
(pi_env) sunteng@zixiang:~/isaacgym$ cd python/
(pi_env) sunteng@zixiang:~/isaacgym/python$ pip install -e .
```

10. 测试环境安装是否成功：

- 先输入： `cd examples`
- 然后输入 `python 1080_balls_of_solitude.py`

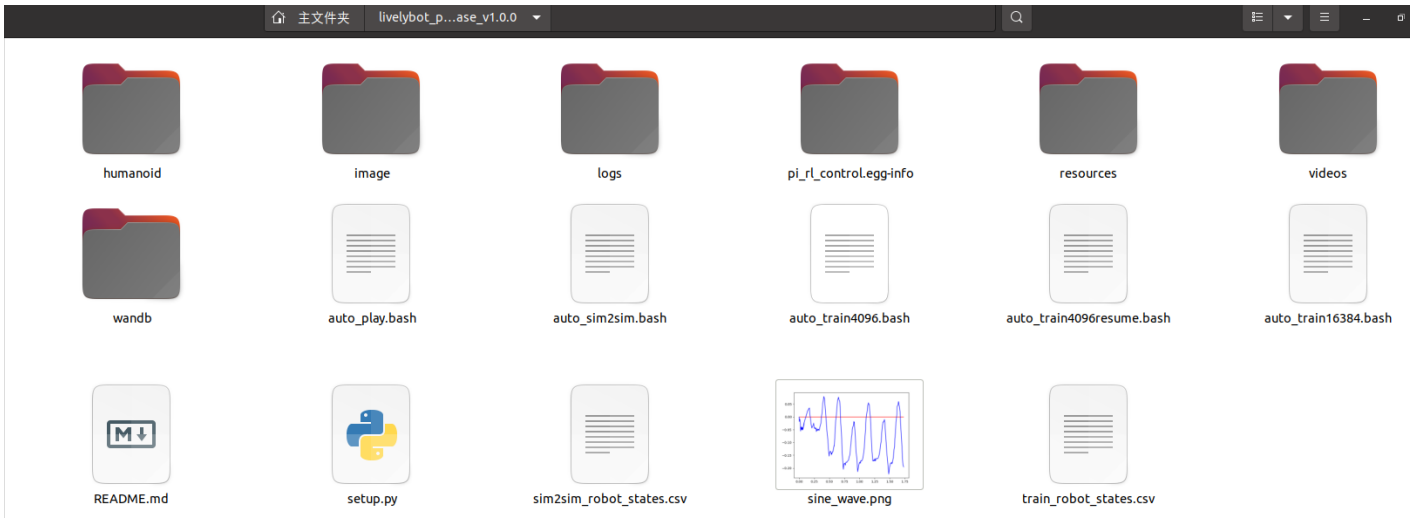
```
(pi_env) sunteng@zixiang:~/isaacgym/python$ cd examples/
(pi_env) sunteng@zixiang:~/isaacgym/python/examples$ python 1080_balls_of_solitude.py
```

如看到下图所示页面，则说明环境安装成功。



11. 下载pi_rl_baseline的zip包：

下载成功后将提取到主目录下，点进livelybot_pi_rl_baseline-release_v1.0.0（1）中，将livelybot_pi_rl_baseline-release_v1.0.0剪切到主目录下；



12. 以开发模式安装当前项目：

- 先输入 `cd livelybot_pi_rl_baseline-release_v1.0.0`
- 再输入 `pip install -e .` *（注意：这里的'.'一定要加上）*

```
(pi_env) sunteng@zixiang:~$ cd livelybot_pi_rl_baseline-release_v1.0.0/
(pi_env) sunteng@zixiang:~/livelybot_pi_rl_baseline-release_v1.0.0$ pip install -e .
```

最后会出现**successfully installed**一系列依赖配置成功。

13. 使用 4096 个环境，并以“v1”为训练版本进行 PPO policy 训练：

- 首先输入 `cd humanoid`
- 再次输入 `python scripts/train.py --task=pai_ppo --run_name v1 --headless --num_envs 4096`，如果出现下图所示界面则说明正在以“v1”为训练版本进行 PPO policy 训练。

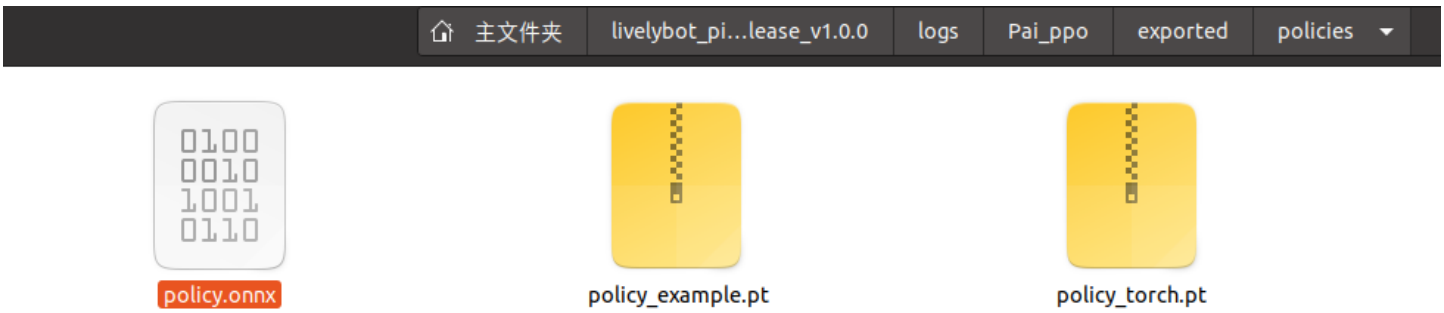
```
wandb: Currently logged in as: 2742974891 (2742974891-beijing-union-university) to https://api.wandb.ai. Use `wandb login --relogin` to force relogin
wandb: Tracking run with wandb version 0.19.11
wandb: Run data is saved locally in /home/sunteng/livelybot_pi_rl_baseline-release_v1.0.0/humanoid/wandb/run-20250603_150509-vpsmk5ib
wandb: Run `wandb offline` to turn off syncing.
wandb: Syncing run Jun03_15-05-08_Pai_ppo_v1
wandb: ⭐ View project at https://wandb.ai/2742974891-beijing-union-university/pai_ppo
wandb: 🚀 View run at https://wandb.ai/2742974891-beijing-union-university/pai_ppo/runs/vpsmk5ib
wandb: WARNING Found log directory outside of given root_logdir, dropping given root_logdir for event file in /home/sunteng/livelybot_pi_rl_baseline-release_v1.0.0/logs/Pai_ppo/Jun03_15-05-08_v1
################################################################################
                      Learning iteration 0/10001

                               Computation: 31499 steps/s (collection: 2.930s, learning 0.190s)
                      Value function loss: 0.3565
                           Surrogate loss: -0.0018
                    Mean action noise std: 1.00
                              Mean reward: 2.02
                      Mean episode length: 23.09
    Mean episode rew_action_smoothness: -0.0009
            Mean episode rew_base_acc: 0.0000
         Mean episode rew_base_height: 0.0069
           Mean episode rew_collision: 0.0000
  Mean episode rew_default_ankle_roll_pos: 0.0044
Mean episode rew_default_hip_roll_joint_pos: 0.0195
 Mean episode rew_default_thigh_joint_pos: 0.0064
             Mean episode rew_dof_acc: -0.0002
             Mean episode rew_dof_vel: -0.0002
       Mean episode rew_feet_air_time: 0.0002
      Mean episode rew_feet_clearance: 0.0139
  Mean episode rew_feet_contact_forces: -0.0001
  Mean episode rew_feet_contact_number: 0.0095
       Mean episode rew_feet_distance: 0.0030
           Mean episode rew_foot_slip: -0.0018
           Mean episode rew_joint_pos: 0.0167
       Mean episode rew_knee_distance: 0.0034
          Mean episode rew_low_speed: -0.0007
        Mean episode rew_orientation: 0.0021
       Mean episode rew_termination: -0.0005
            Mean episode rew_torques: -0.0000
      Mean episode rew_track_vel_hard: -0.0029
    Mean episode rew_tracking_ang_vel: 0.0051
    Mean episode rew_tracking_lin_vel: 0.0180
    Mean episode rew_vel_mismatch_exp: 0.0040
--------------------------------------------------------------------------------
                          Total timesteps: 98304
                          Iteration time: 3.12s
                              Total time: 3.12s
                                     ETA: 31210.8s
```

注意：这里会弹出wandb注册网站进行帐户注册（wandb在Isaac Gym中用于实验跟踪、超参数记录、训练指标可视化的奖励曲线）

14. 评估训练好的policy，自动导出适合部署目的的 JIT 模型：

- 先下载onnx包: `pip install onnx`
- 然后再输入 `python scripts/play.py --task=pai_ppo --run_name v1`，可以在/home/sunteng/livelybot_pi_rl_baseline-release_v1.0.0/logs/Pai_ppo/exported/policies下看到policy.onnx

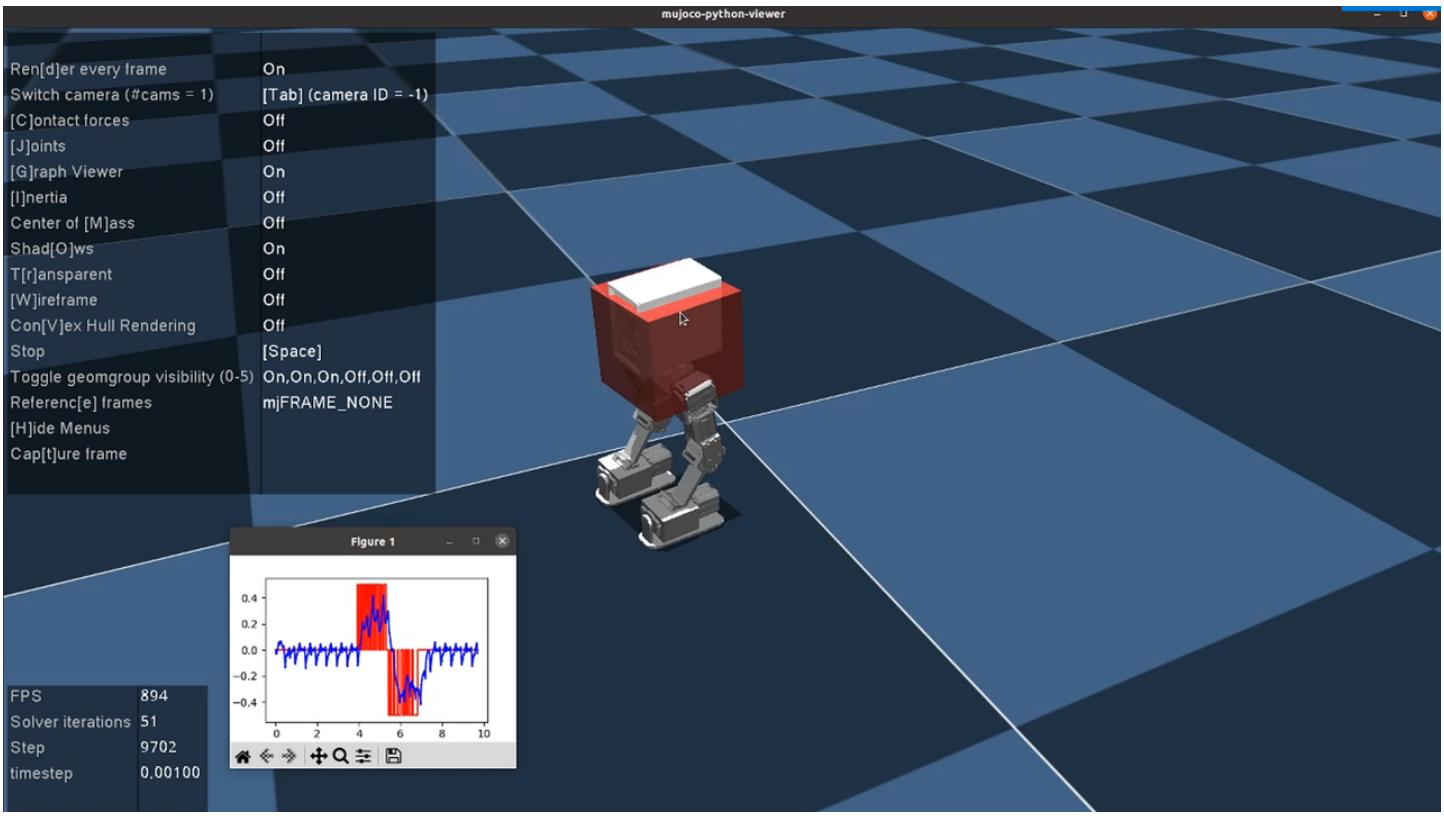


15. 通过使用Mujoco实现sim2sim：

`python scripts/sim2sim.py --load_model /path/to/logs/Pai_ppo/exported/policies/policy_torch.pt` *（注意：这里的/path/to替换为logs上一级文件夹的路径）*

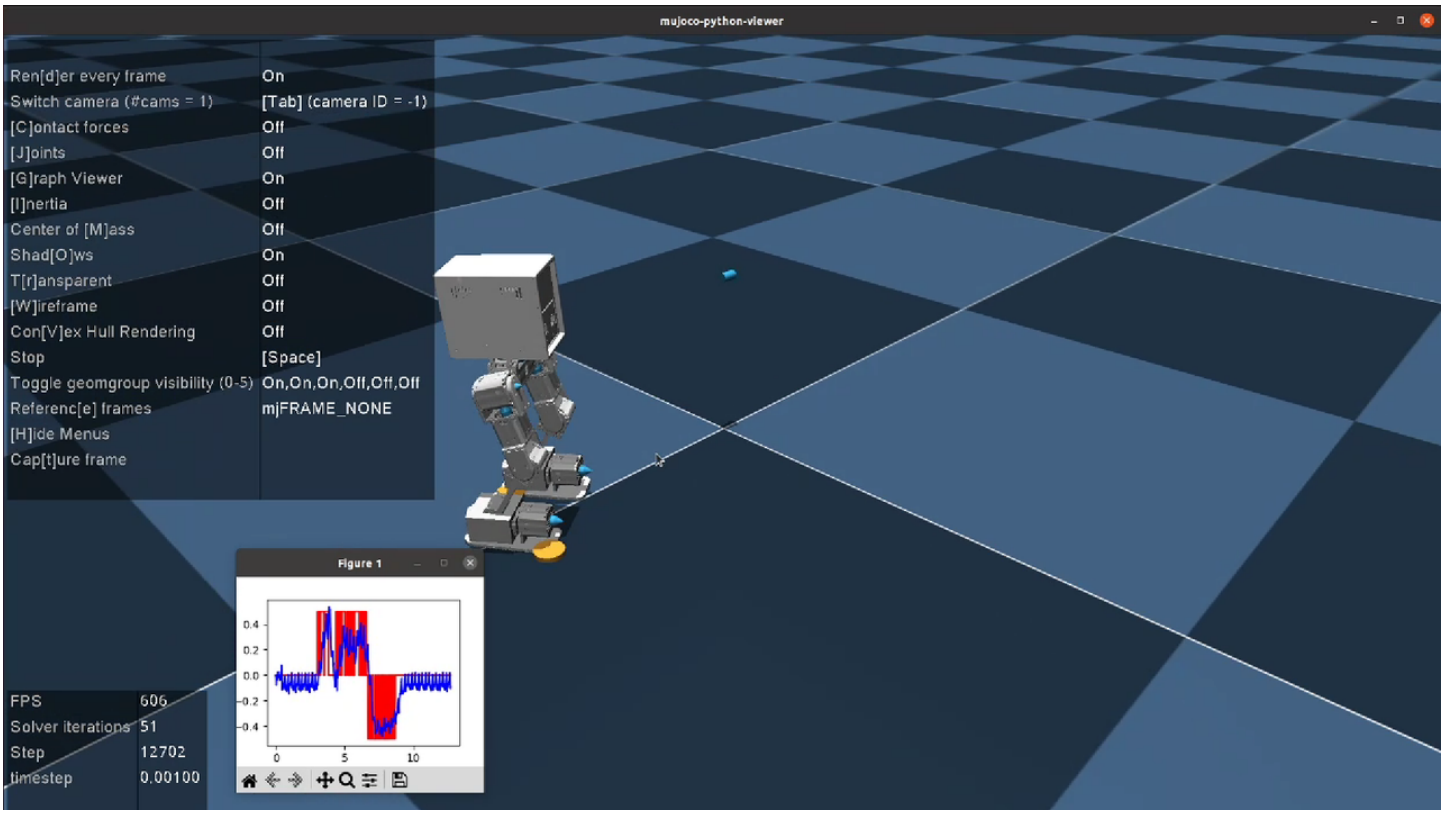


如运行成功，即可看到下图所示mujoco场景：



16. 运行高擎提供的训练好的policy_example：

`python scripts/sim2sim.py --load_model /path/to/logs/Pai_ppo/exported/policies/policy_example.pt` *（注意：这里的/path/to替换为logs上一级文件夹的路径）*



到这里，基本的强化学习baseline教学就结束了，您可以在仿真环境中调试属于自己的机器人。

如需更多技术支持，请致电：+86-18144897433或微信添加：GQJD2022。